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Estimation of Salt-Pepper Noise in Images with Correlation Inspection

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Abstract. We proposed an approach for estimating the density of salt-pepper noise in images with correlation inspection. The correlation coefficients histogram was introduced in this paper to depict the correlation distributions of images. Based on the fact that the correlation distributions of natural images are nearly independent of individual images, we revealed how the correlation coefficients histogram of the noisy image deviates from that of natural image along with the noise density in quantitative form, thus we took advantage of this relation to make estimation. Simulation results showed that the proposed approach outperforms those of existing methods.

Introduction

Noise in images will make image degrade in quality and badly affect the further processings. The salt-pepper noise, which is usually introduced by short-lived halt in imaging, is a typical noise type. Such noise not only disturbs our visual sense, but it also misleads many operations, e.g., edge detection. Therefore filtering of salt-pepper noise in images is vital for the entire image processing procedure, this preprocess is called denoising in the image processing literatures.

At present, the median filtering, as a nonlinear processing, has become one of most effective algorithms for filtering of salt-pepper noise. Many other improved algorithms based on it have been developed by numerous scholars and researchers in recent years, for example, the adaptive median filter, the adaptive fuzzy switching filter, the threshold switching median filter, the progressive switching median filter, the filter with extremum and median values, the weighted median filter, the filter using long-range correlation, etc[1-8]. It was found that most of these algorithms need to take advantage of the parameter of the noise density in operations, and usually they assume that the parameter is known, or make partial estimation along with denoising, with which the perfect outcomes can not be achieved in general. The noise density is usually unknown in reality, thus making the estimation of salt-pepper noise becomes a groundwork for denoising and the precision of the estimation value may directly affect the performance of denoising. Furthermore, the estimation of noise also helps to make strategy for image processing. For example, if we blindly apply the strong filters on a image with little noise, we are likely to get the more degraded image, instead of the improved one. But if we are aware of the noise level by estimation, then we know that here the slight filtering or non-denoising is appropriate.

Now the estimation of noise has become a research hotspot in the field of denoising and a great deal of papers with such topic have been reported in recent years. However, the estimation algorithms for salt-pepper noise were seldom appeared in publications while most of those aimed at Gaussian noise. Lately Zhang and Chao et al. have searched after the estimation of salt-pepper noise and made great progress in it[9-10]. Zhang et al. [9] presented an estimation algorithm benefiting from the fact that the variance of wavelet coefficients in the diagonal subband is related to the density of the salt-pepper noise. Chao et al. [10] proposed another method using the relation between the magnitude spectrum of the noisy image and the density of the salt-pepper noise. Both algorithms can lead to better results, but the estimation values of both ones are still not adequately precise, which may be incapable of meeting some denoising algorithms sensitive to the noise parameters.

To enhance the precision of the estimation values, in this paper we proposed a novel algorithm based on the correlation inspection(called the proposed approach hereinafter). We above all introduced the correlation coefficients histogram(CCH for short) to depict the entire correlation distribution of a image, and exhibited that the CCHs of natural images are independent of individual image traits. Based on it, we further revealed that the CCH of the noisy image monotonously deviates from that of the original image along with the density of the salt-pepper noise in the image. Such change relation has strong robusticity, that is, it is nearly foreign to individual image, thus we took advantage of this relation to make estimation. Simulation results indicated that the proposed approach can achieve better outcomes than those of the methods cited above.

The rest of the paper is organized as follows. Section 2 summarizes the correlation distributions of natural images and analyses the noise impact on the CCH. We present the estimation algorithm in section 3. Numerical results and performance evaluation are reported in section 4. We conclude this paper in section 5.

The Correlation Distribution of the Image

In general the salt-pepper noise has much more remarkable magnitudes than the image signal, so the gray levels of the noise points are often digitalized as the extremums of entire grayscale area. In the case of 8-bits presentation, the extremums are 0 and 255, contributed by the pepper noise and the salt noise respectively. Usually both noise components have the same weights in quantity. Therefore, if the density of the noise is φ, we can model the salt-pepper noise with Eq. 1:

$$g(x,y)=\begin{cases}0, & P(0)=\varphi/2\\ 255, & P(255)=\varphi/2\\ f(x,y), & P(f(x,y))=1-\varphi\end{cases}\quad. \tag{1}$$

where $f(x,y)$ and $g(x,y)$ denote the grayscale values of the pixels (x,y) in the original image and the noisy image, respectively. $P(x)$ denotes the probability that the value gets x. It implies that about $N\times\varphi$ pixels are polluted by noise if the total number of the image pixels is N.

CCHs of Natural Images. For natural images, the neighboring pixels are strongly correlative[11]. Usually the pixel points have the same or close gray levels as other pixels in the neighborhoods. In order to measure the correlation of a pixel point with the neighboring pixels, we introduced the correlation coefficient of an individual pixel. Let (a,b) be the location of the inspected pixel, and $f(x,y)$ be the grayscale value of the pixel located at (x,y). The 8-neighborhood of the pixel (a,b) is formulated as:

$$g1(x',y')=f(x',y')\,. \tag{2}$$

where $a-1\le x'\le a+1, b-1\le y'\le b+1$. We define the template of the pixel (a,b) as Eq. 3:

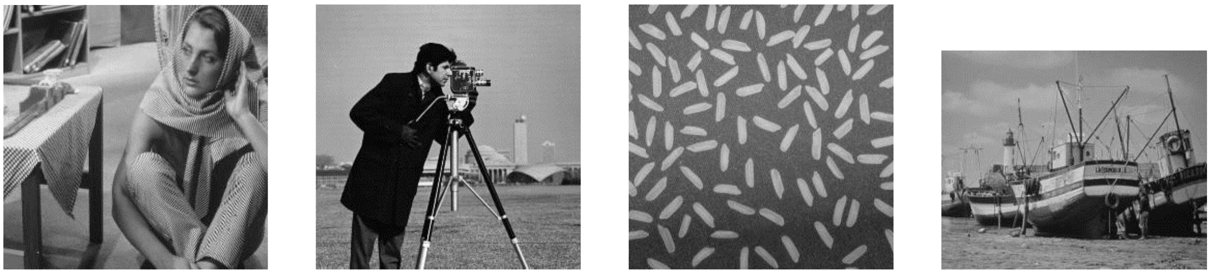
$$g2(x',y')=f(a,b)\,. \tag{3}$$

where x' and y' satisfy $a-1\le x'\le a+1, b-1\le y'\le b+1$, too, that is, $g1$ and $g2$ have the same sizes. Note that each pixel in the template has the same grayscale value as the pixel (a,b) in the image. Therefore the correlation coefficient of the pixel (a,b) can be defined using Eq. 4:

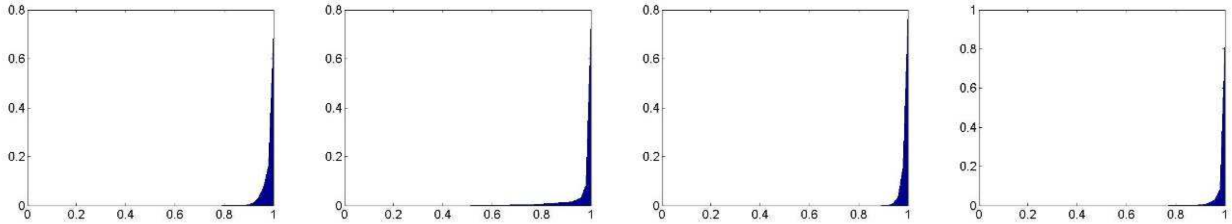
$$\rho(a,b)=\frac{\sum_{x'}\sum_{y'}g1(x',y')\times g2(x',y')}{\sqrt{\sum_{x'}\sum_{y'}[g1(x',y')]^2\times\sum_{x'}\sum_{y'}[g2(x',y')]^2}}\,. \tag{4}$$

If the pixel (a,b) has complete correlation with other 8 pixels in the neighborhood, $\rho(a,b)$ gets the value 1 due to $g1$ and $g2$ being the same. While the correlation gets degraded, $\rho(a,b)$ becomes small up to zero. So Eq. 4 indicates the correlation of the pixel (a,b) with the neighboring pixels rightly. In scenarios where the gray level is stored with 8 bits, as we apply Eq. 4 on the black pixel, which has the grayscale value 0, the denominator will be zero and it will result in overflow problem. To avoid such exception, we represent the image in the double form and enhance the entire image by one scale unit, and this operation will not virtually impact the correlation results.

With corresponding $g1$ and $g2$, we apply Eq. 4 on all image pixels to get each correlation coefficient value, then we get another image, called the correlation coefficient image(CCI for short), which has the same size as the original image. The value area of CCI is [0 1] and the pixel value here denotes the correlation of the corresponding pixel in the original image with the neighboring pixels. To make out the correlation distribution of the original image, we make the histogram of the CCI, such histogram is called the correlation coefficients histogram(CCH for short). For example in Fig. 1, figures e, f, g and h are the normalized CCHs of four natural images labeled as a, b, c and d, respectively. Where horizontal axis denotes the correlation coefficient, and vertical one denotes the percentage ratio of each statistical section.



(a) Barbara (b) Cameraman (c) Rice (d) Ship



(e) CCH of Fig. a (f) CCH of Fig. b (g) CCH of Fig. c (h) CCH of Fig. d

Fig. 1 Original images and their CCHs

It is clear that here all four CCHs have very close forms. Experimental results on a great lot of other images showed that all their CCHs also conform to such distribution. It indicated that for natural images, most pixels have strong correlations with other pixels in the neighborhoods, and it's more notable, the correlation distributions of natural images are stable, which hardly change with the individual images. It implies that the CCH is robust to image traits and it is a proper candidate for the universal characteristic of all natural images. To gain more robust outcome, we made average of a lot CCHs from different images and got the standard CCH shown in Fig. 2. Now the standard CCH can be used in any circumstance as an apriori knowledge, no matter what the individual image is.

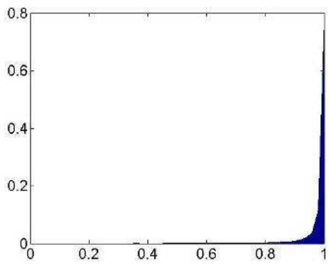
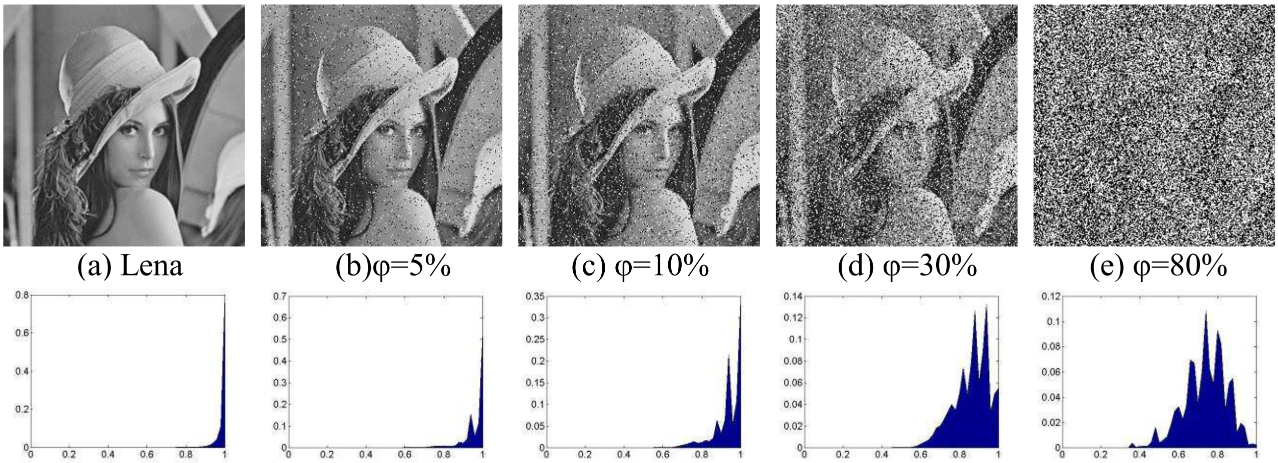


Fig. 2 Standard CCH

Noise Impact on the CCH. An original image not with noise will become a noisy image when it is polluted by salt-pepper noise. The noisy image contains some noise pixels which have grayscale extremums. The imposed noise pixels, which are not the significative components of the original image, usually have slight or no correlations with the neighboring pixels contributed by the original image. For these noise points, thus the correlation coefficients attained using Eq. 4 are small in general. The larger the noise density becomes, the more the pixels turn into noise points. Therefore we conclude that when the noise density becomes larger and larger, the pixels with small correlation coefficients get more and more, and on the other hand the ones with large correlation coefficients get less and less. In other words, the CCH of the noisy image deviates from that of the original image (or the standard CCH) more and more while the density of the salt-pepper noise becomes larger and larger. To verify such conclusion, we took a great deal of images and dirtied them with each level noise. We made CCHs for all these noisy images and results indicated that our conclusion is completely just for all images. For instance, Fig. 3 shows the noise impact on the CCH. Images and noise densities are shown at left side, and right side are the corresponding CCHs. Here φ denotes the density of the salt-pepper noise.



(a) Lena (b)φ=5% (c) φ=10% (d) φ=30% (e) φ=80%

(f) CCH of Fig. a (g) CCH of Fig. b (h) CCH of Fig. c (i) CCH of Fig. d (j) CCH of Fig. e

Fig. 3 Demonstration of the noise impact on the CCH

It is clearly shown in Fig. 3 that the more the noise pixels become, the more the CCH of the noisy image deviates from that of the original image (or the standard CCH), and this deviation is sensitive to the noise density. For example, 5% and 10% are close values, however the obtained CCHs are remarkably dissimilar. Note that for all other images, this relation comes into existence, too. It means that we can take advantage of such relation to estimate the density of the salt-pepper noise. In order to make this relation formalized and usable, we will quantitate the deviation. Let $hist(1:H)$ be the data of the standard CCH, where H denotes the length of the vector $hist$, it implies that when we make histogram of the CCI, we divide the entire value area into H statistical parts equably. Similarly, $nohist(1:H)$ denotes the CCH of the noisy image. Note that $nohist$ has the same length as $hist$. $hist(i)$ is the percentage ratio of those values standing in the i th section, and $nohist(i)$ denotes similar meanings. Therefore we can formulate the deviation using the correlation coefficient between both CCHs, defined as Eq. 5:

$$\rho(hist, nohist) = \frac{\sum_{i=1}^{H} hist(i) \times nohist(i)}{\sqrt{\sum_{i=1}^{H} [hist(i)]^2 \times \sum_{i=1}^{H} [nohist(i)]^2}} . \quad (5)$$

If $nohist$ is the same as $hist$, $\rho(hist, nohist)$ gets the value 1. While $nohist$ deviates from $hist$ more and more, $\rho(hist, nohist)$ becomes less and less, up to zero. So Eq. 5 rightly depicts the deviation between both CCHs.

Now we imposed the salt-pepper noises on all five original images in Fig. 1 and 3 to make various noisy images with the noise densities of 5%, 10%, 15%, 20%, 25%, 30%, 35% and 40% respectively, and then got the correlation coefficient between the standard CCH and the CCH of each noisy image with Eq. 4 and Eq. 5. Results data are shown in Table 1. According to these data, we drew five corresponding curves in Fig. 4 to show the noise impact on the correlation coefficient $\rho(hist, nohist)$. Where horizontal axis denotes the noise density, and vertical one denotes the value of $\rho(hist, nohist)$.

Table 1 Experimental data

Noise density	Correlation coefficient $\rho(hist, nohist)$				
	Fig. 1a	Fig. 1b	Fig. 1c	Fig. 1d	Fig. 3a
5%	0.9421	0.9599	0.9526	0.9648	0.9575
10%	0.8160	0.8431	0.8138	0.8389	0.8367
15%	0.6500	0.6556	0.6178	0.6568	0.6771
20%	0.4895	0.5223	0.4406	0.4796	0.5228
25%	0.3419	0.3545	0.3325	0.3370	0.3714
30%	0.2397	0.2408	0.2027	0.2251	0.2606
35%	0.1763	0.1734	0.1420	0.1487	0.1846
40%	0.1249	0.1142	0.0991	0.1006	0.1250

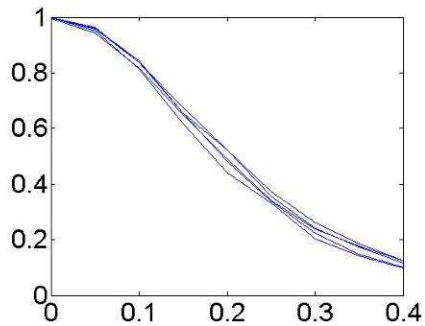


Fig. 4 Curves showing the noise impact on the correlation coefficient $\rho(hist, nohist)$

We made the simulation on lots of other images and achieved similar outcomes. Based on the analysis of all these data, we summarized the results with next two verdicts:

(1)The correlation coefficient $\rho(hist, nohist)$ between the standard CCH and the CCH of the noisy image monotonously diminishes along with the increase of the noise density.

(2)The function relation between $\rho(hist, nohist)$ and the noise density is nearly foreign to individual images. The value of $\rho(hist, nohist)$ is determined almost only by the noise density. It can be drawn from the fact that all five curves in Fig. 4 overlap together on the whole.

Above analyses revealed that the correlation coefficient $\rho(hist, nohist)$ can be used to estimate the noise density and may lead to trusty outcomes. To make such relation between $\rho(hist, nohist)$ and the noise density more robust, we drew statistical results from much simulation data. Part statistical data on some noise densities are shown in Table 2 and the corresponding curve is drawn in Fig. 5.

According to the statistical data, we made polynomial fit by the rule of minimum mean square error(MMSE). The gained formula is shown as Eq. 6:

$$\varphi(\rho) = 54.93 - 124.91 \times \rho + 153.34 \times \rho^2 - 81.77 \times \rho^3 . \tag{6}$$

where ρ denotes the value of $\rho(hist, nohist)$ obtained using Eq. 5 and φ denotes the density of the salt-pepper noise. Here if φ gets the value γ, it means that the estimation value of the noise density is $\gamma\%$. Eq. 6 represents the relation between $\rho(hist, nohist)$ and the noise density in quantitative form. In scenarios where the precision is not required strictly, we may make two-order fit.

Table 2 Part statistical data

Noise density	Statistical value of $\rho(hist, nohist)$
5%	0.9424
10%	0.8323
15%	0.6802
20%	0.5238
25%	0.3875
30%	0.2796
35%	0.2016
40%	0.1457

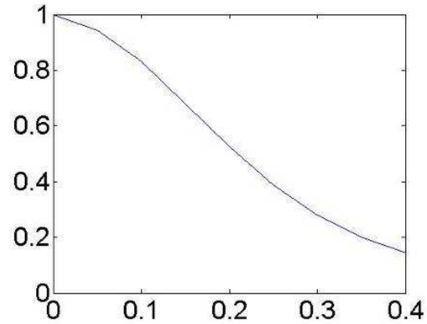


Fig. 5 Curve showing the statistical data

Estimation of Noise with Correlation Inspection

Section 2 revealed how the correlation coefficient between the standard CCH and the CCH of the noisy image changes along with the noise density in quantitative form by Eq. 6. Experiment results indicated that such relation is robust to individual image traits. In this paper we took advantage of this relation to estimate the noise density using the standard CCH and Eq. 6 as apriori knowledge, and achieved better performances than those of existing algorithms.

Estimation of the salt-pepper noise in a noisy image can be summarized with the following steps:

(1)For each pixel in the noisy image, get it's correlation coefficient value using Eq. 4 and then make the correlation coefficient image(CCI).

(2)Make histogram of the CCI and then get the correlation coefficients histogram(CCH) of the noisy image.

(3)With Eq. 5, get the correlation coefficient between the standard CCH and the CCH of the noisy image.

(4)With the correlation coefficient value gained in step (3), we get the final result using Eq. 6, which denotes the estimation value of the noise density.

Note that when we make CCH of the noisy image, the statistical manner will be the same as the manner in making the standard CCH. It implies that the lengths of both CCHs are the same. In experiments we found that the length of CCH, which represents the number of the divided sections for statistic, can not be too large or too small. Improper length values will lead to imperfect outcomes. It was found from much simulation data that the appropriate value should fall into [40 60]. In this paper we took the value 50, and note that the data and conclusions in the previous section were gained based on this value, too.

Simulation Results and Performance Evaluations

We executed the simulation in Matlab7.0 to verify the proposed approach. We took 30 natural images randomly and dirtied them with the salt-pepper noises, whose densities are 5%, 10%, 15%, 20%, 25%, 30%, 35% and 40% respectively. For each noisy image, we estimated the noise density with the proposed approach and the estimation results are shown in Table 3. To compare with existing methods, outcomes resulting from other two algorithms in literature [9] and [10] are shown in the same table. In table 3, φ is the truth value of the noise density, $mean(\hat{\varphi})$ is the mean value of 30 estimation results, and $std(\hat{\varphi})$ is the standard deviation from $mean(\hat{\varphi})$. Note that all 30 original images here are different from those ones used to gain apriori knowledge in section 2, we did so to indicate the robusticity of our algorithm. Some of experimental images are shown in Fig. 6.

The data in Table 3 indicates that the proposed approach outperforms existing methods. Only on the truth value of 5% our algorithm does worse than the one in [9]. At any other instance, the proposed approach can achieve more precise outcomes than other two methods. When we employ the method in [9], the estimation result deviates from the truth value more and more while the noise density becomes larger and larger. For the algorithm in [10], on the contrary, the smaller the noise density becomes, the more the estimation value deviates from the truth one. These two specialities do not stand in our approach, that is, the deviation from the truth value does not change along with the noise density in the case of the proposed approach. It indicates that the relation between the correlation coefficient $\rho(hist, nohist)$ and the noise density used in our algorithm is more robust than those of affiliations in other methods. As for standard deviation of the estimation results, the performance of our approach is close to that of the method in [9]. The method in [10] can lead to smaller standard deviation values on the whole, just meaning that the fluctuation of it's estimation values is less than that of our results. However note that here mean value is much more significant than standard deviation. The outcomes which fluctuate more but approach the truth value are more acceptable in comparison with the results which fluctuate slightly but are far from the truth value.

Table 3 Estimation results of three algorithms

φ	Proposed approach		Method in [9]		Method in [10]	
	$mean(\hat{\varphi})$	$std(\hat{\varphi})$	$mean(\hat{\varphi})$	$std(\hat{\varphi})$	$mean(\hat{\varphi})$	$std(\hat{\varphi})$
5	4.3951	1.5287	4.9568	0.8161	4.1410	2.0487
10	9.6544	1.8855	9.6437	1.4449	11.305	1.8805
15	14.937	1.9332	14.281	1.9504	16.488	1.7133
20	19.838	2.4343	19.020	2.7044	21.978	2.1352
25	24.915	3.2048	23.627	3.2303	25.223	2.6145
30	30.219	4.0153	28.316	3.6841	29.076	3.0581
35	35.353	4.4589	33.217	4.2381	34.643	3.5397
40	39.678	4.3465	37.993	4.5405	40.925	3.7142

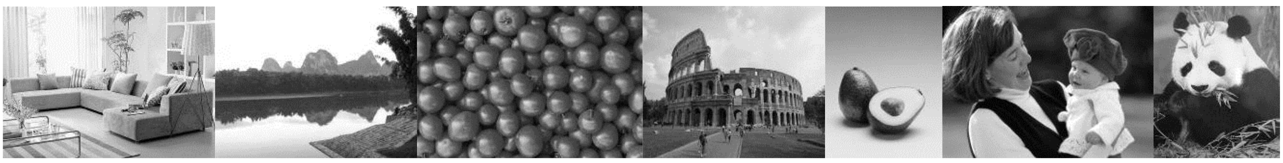


Fig. 6 Some of experimental images

Conclusions

In this paper we researched the correlation distributions of natural images and indicated that for all natural images, their CCHs have similar forms. We further revealed that the CCH of the noisy image deviates from the standard CCH more and more while the density of the salt-pepper noise becomes larger and larger, and we finally represented this relation in quantitative form. We took advantage of such relation to estimate the noise density and achieved better results than those of existing algorithms.

The proposed approach exhibits better performance when the noise density is not too large(<60%), but the performance will degrade in the case of high level noise(>70%). It can be explained that when the noise density becomes very large, the noise points congregate close, here the correlation coefficients of the noise pixels may become larger instead of smaller because their neighborhoods may contain some other noise points, which have the same grayscale values as the inspected pixels with probability of 50%. Thus the CCH of the noisy image no longer monotonously deviates from the standard CCH along with the noise density and so it leads to imprecise results. Such problem also lies in other algorithms including both ones cited above. Improving the performance in the case of high level noise is our next-step pursuit. However, the density of the salt-pepper noise in images is not too large in general, so our approach can be properly used in practice and acts as an important role.

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